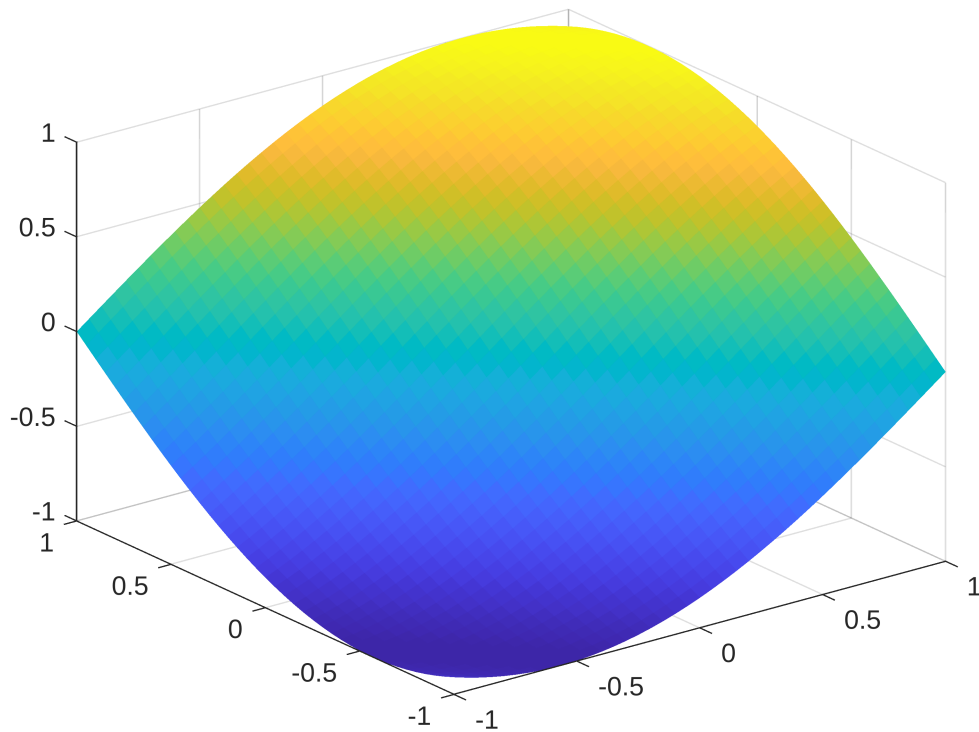


Practice Qn's

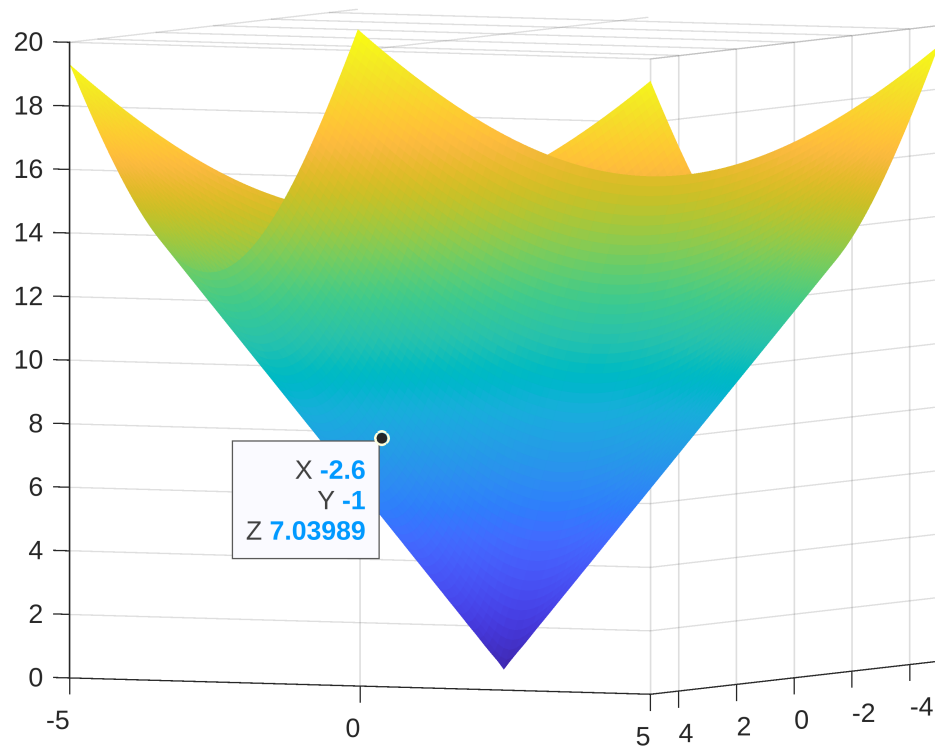
Question 1

```
syms x y z;  
[X,Y] = meshgrid(-1:0.05:1,-1:0.05:1);  
Z = sin(X+Y);  
surf(X,Y,Z,'EdgeColor','none');
```



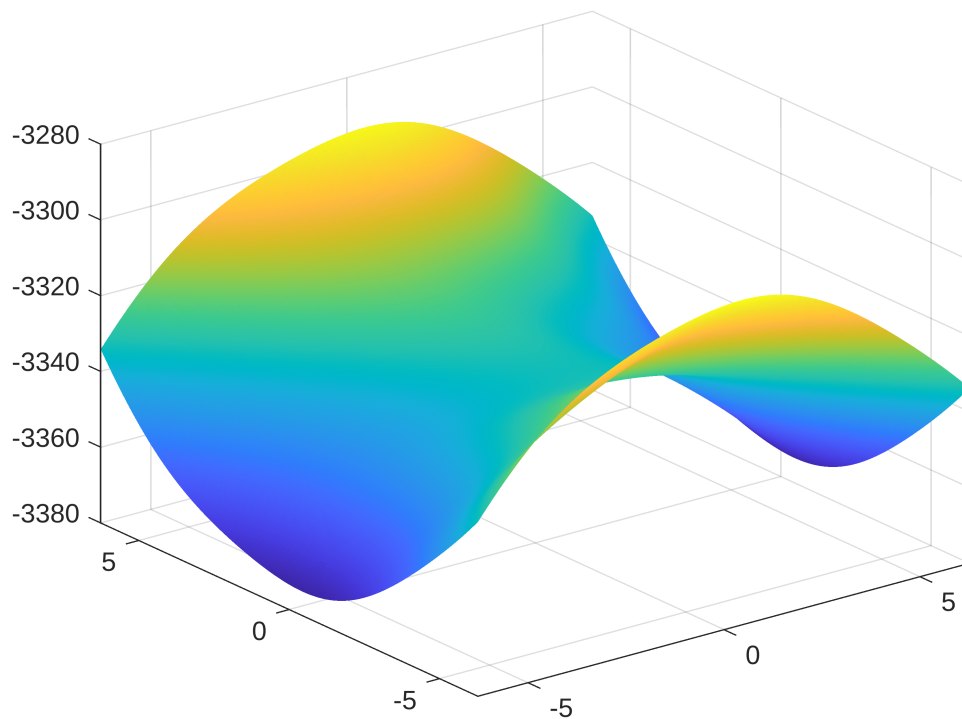
Question 2

```
[X,Y] = meshgrid(-5:0.1:5,-5:0.1:5);  
Z = sqrt(abs(6*(X.^2)+9*(Y.^2)));  
surf(X,Y,Z,'EdgeColor','none');
```



Question 3

```
[X,Y] = meshgrid(-2*pi:0.05:2*pi,-2*pi:0.05:2*pi);  
Z = sin(X+Y) - X*Y - X.^2 + Y.^2;  
surf(X,Y,Z, 'EdgeColor', 'none');
```

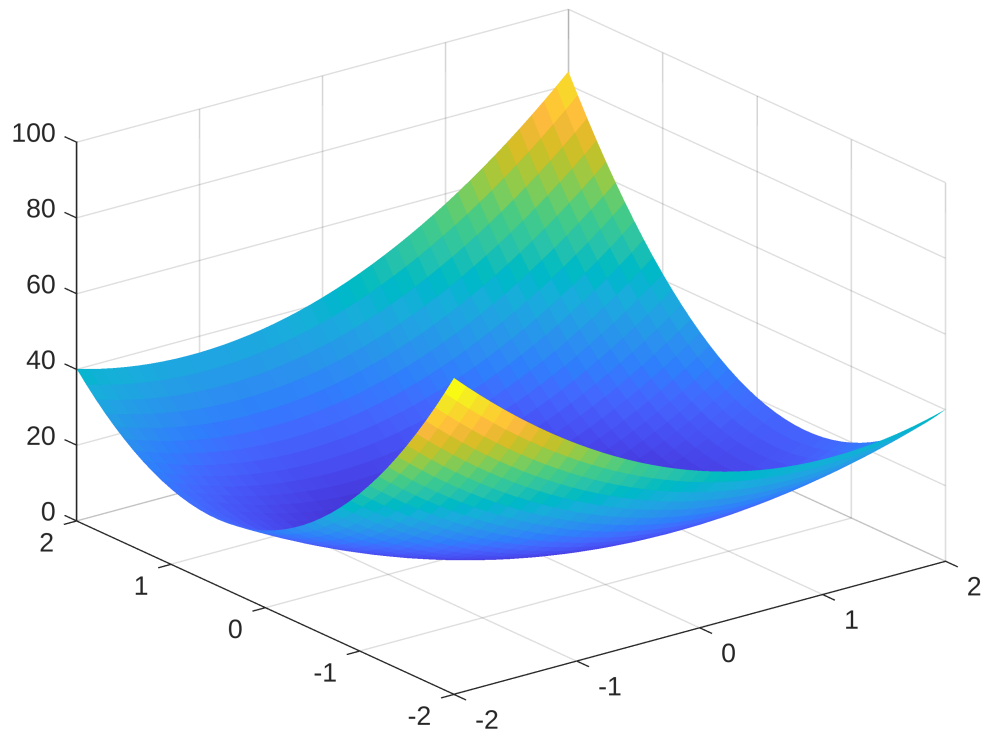


Question 4

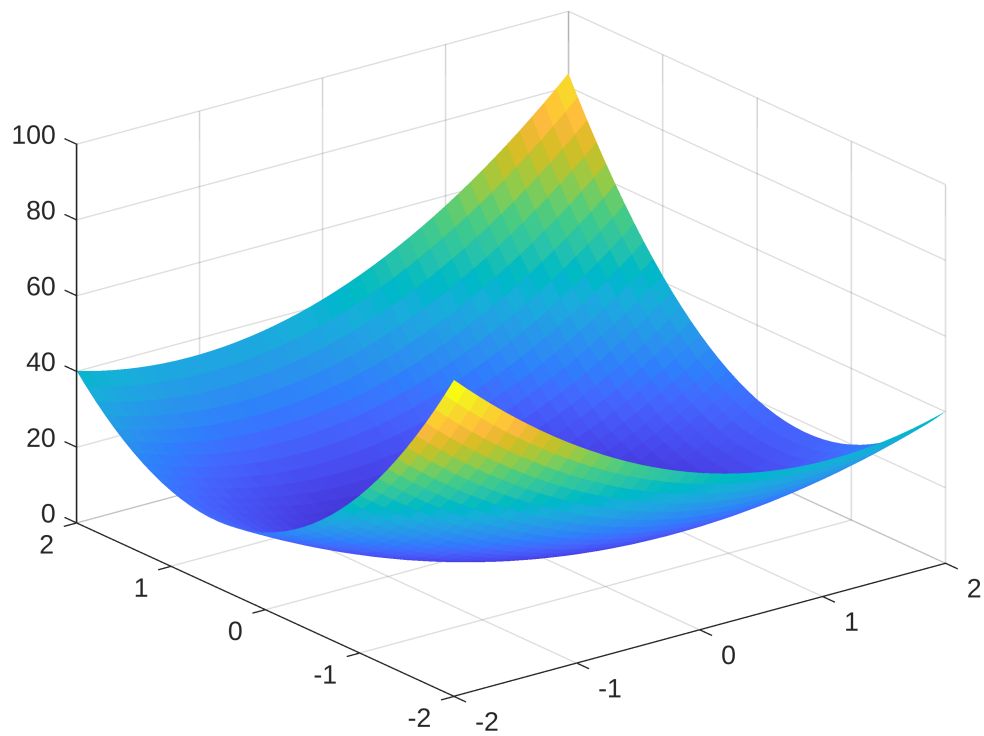
```
A = [3 2;
     2 5];
B = [9 4;
     4 6];
C = [-5 3;
     3 8];
D = [-1 -1;
     -1 -4];

[x1, x2] = meshgrid(-2:0.1:2, -2:0.1:2);
f = S(1,1)*x1.^2 + 2*S(1,2)*x1.*x2 + S(2,2)*x2.^2;

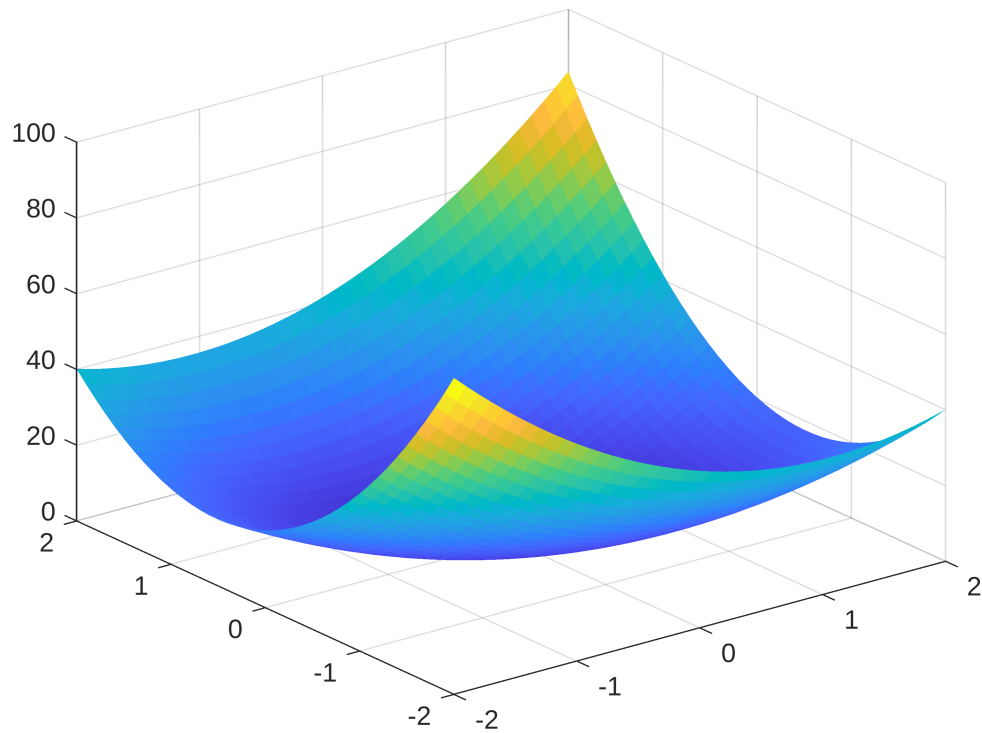
S = A;
surf(x1, x2, f, 'EdgeColor', 'none');
```



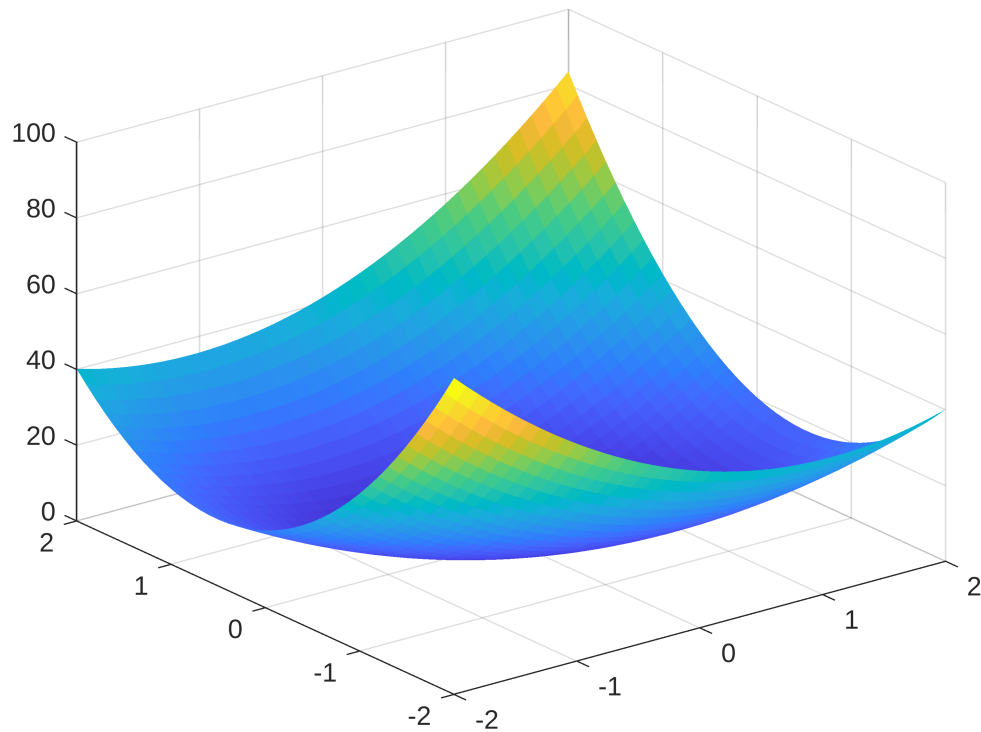
```
S = B;  
surf(x1, x2, f, 'EdgeColor', 'none');
```



```
S = C;  
surf(x1, x2, f, 'EdgeColor', 'none');
```



```
S = D;  
surf(x1, x2, f, 'EdgeColor', 'none');
```



Question 5

```
A = randn(6, 2);
S = A' * A;
```

% Part (a): Find eigenvalues of S to check positive definiteness

```
eigenvalues = eig(S);
disp('Eigenvalues of S:');
```

Eigenvalues of S:

```
disp(eigenvalues);
```

```
3.9562
6.4126
```

% Part (b): Plot the quadratic form $f(x) = x' * S * x$

```
[x, y] = meshgrid(-1:0.1:1, -1:0.1:1);
```

```
f = zeros(size(x));
```

```
for i = 1:size(x,1)
```

```
    for j = 1:size(x,2)
```

```
        v = [x(i,j); y(i,j)];
```

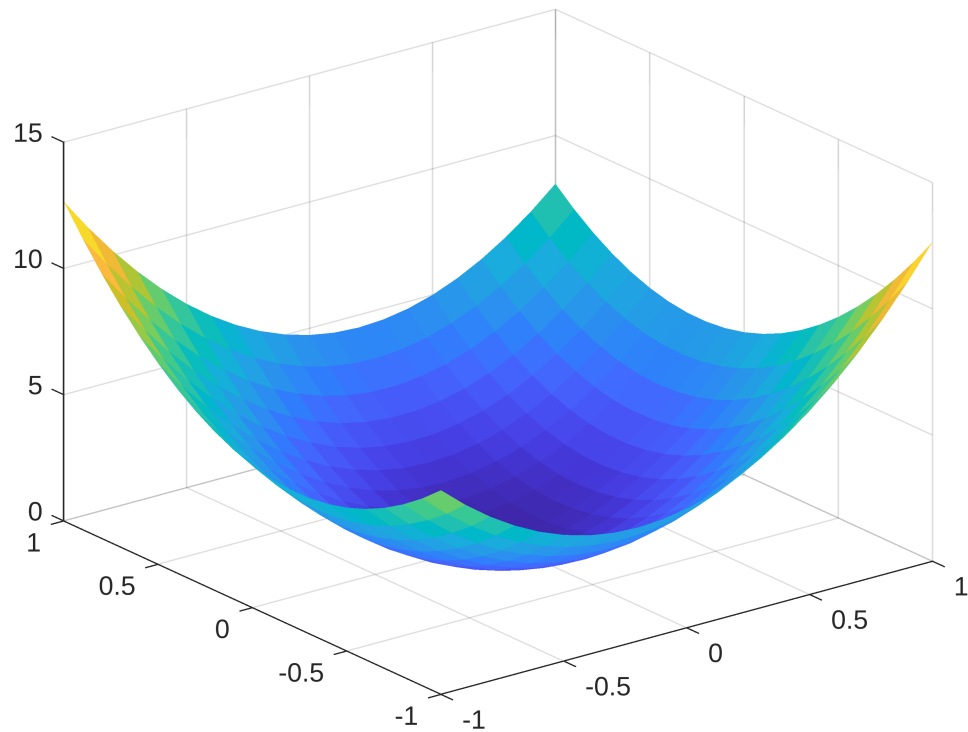
```
        f(i,j) = v' * S * v;
```

```
    end
```

```
end
```

```
figure;
```

```
surf(x, y, f, 'EdgeColor', 'none');
```



Question 6

```
syms x y z
```

% Part (a)

```
f = 4*x^3*exp(x*y) - y^3*z;  
grad_f = gradient(f, [x, y, z])
```

grad_f =

$$\begin{pmatrix} 12x^2e^{xy} + 4x^3ye^{xy} \\ 4x^4e^{xy} - 3y^2z \\ -y^3 \end{pmatrix}$$

```
hess_f = hessian(f, [x, y, z])
```

hess_f =

$$\begin{pmatrix} 24 x e^{x y} + 24 x^2 y e^{x y} + 4 x^3 y^2 e^{x y} & \sigma_1 & 0 \\ \sigma_1 & 4 x^5 e^{x y} - 6 y z & -3 y^2 \\ 0 & -3 y^2 & 0 \end{pmatrix}$$

where

$$\sigma_1 = 16 x^3 e^{x y} + 4 x^4 y e^{x y}$$

% Part (b)

```
g = exp(x*y)*(x + y);
grad_g = gradient(g, [x, y]);
hess_g = hessian(g, [x, y]);
grad_g_at_P = subs(grad_g, [x, y], [-1, 1])
```

grad_g_at_P =

$$\begin{pmatrix} e^{-1} \\ e^{-1} \end{pmatrix}$$

```
hess_g_at_P = subs(hess_g, [x, y], [-1, 1])
```

hess_g_at_P =

$$\begin{pmatrix} 2 e^{-1} & 0 \\ 0 & -2 e^{-1} \end{pmatrix}$$

% Part (c)

```
h = (x^2 + y^2)/(x^2 - y^2);
grad_h = gradient(h, [x, y, z])
```

grad_h =

$$\begin{pmatrix} \frac{2 x}{x^2 - y^2} - \frac{2 x (x^2 + y^2)}{(x^2 - y^2)^2} \\ \frac{2 y}{x^2 - y^2} + \frac{2 y (x^2 + y^2)}{(x^2 - y^2)^2} \\ 0 \end{pmatrix}$$

```
hess_h = hessian(h, [x, y, z])
```

hess_h =

$$\begin{pmatrix} \frac{2}{x^2 - y^2} - \frac{2\sigma_4}{\sigma_1} - \frac{8x^2}{\sigma_1} + \frac{8x^2\sigma_4}{\sigma_3} & \sigma_2 & 0 \\ \sigma_2 & \frac{8y^2}{\sigma_1} + \frac{2\sigma_4}{\sigma_1} + \frac{2}{x^2 - y^2} + \frac{8y^2\sigma_4}{\sigma_3} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

where

$$\sigma_1 = (x^2 - y^2)^2$$

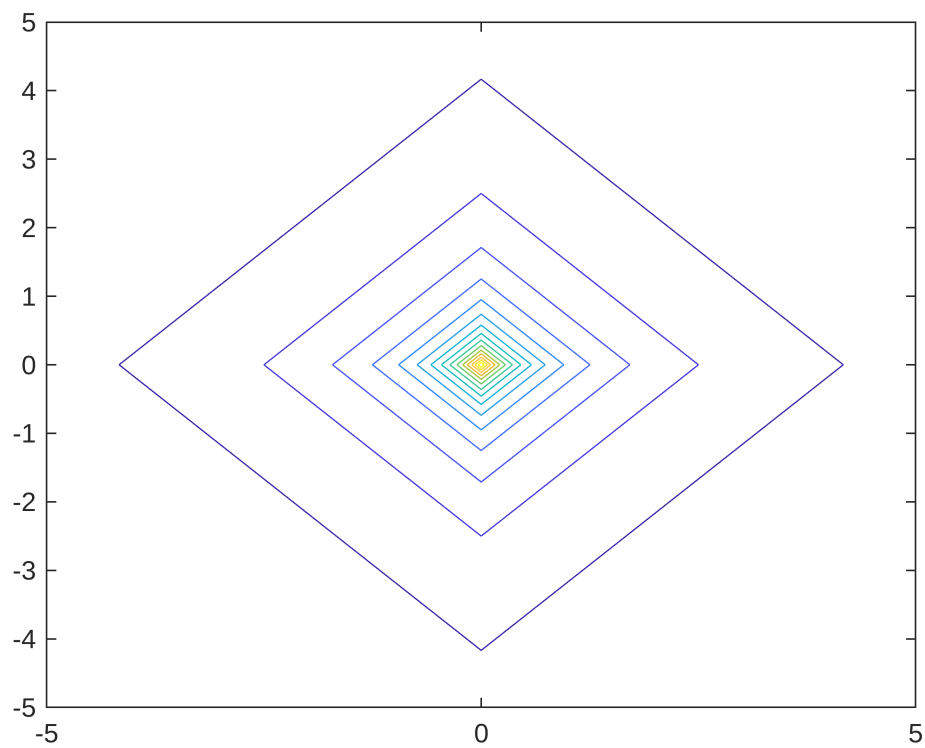
$$\sigma_2 = -\frac{8xy\sigma_4}{\sigma_3}$$

$$\sigma_3 = (x^2 - y^2)^3$$

$$\sigma_4 = x^2 + y^2$$

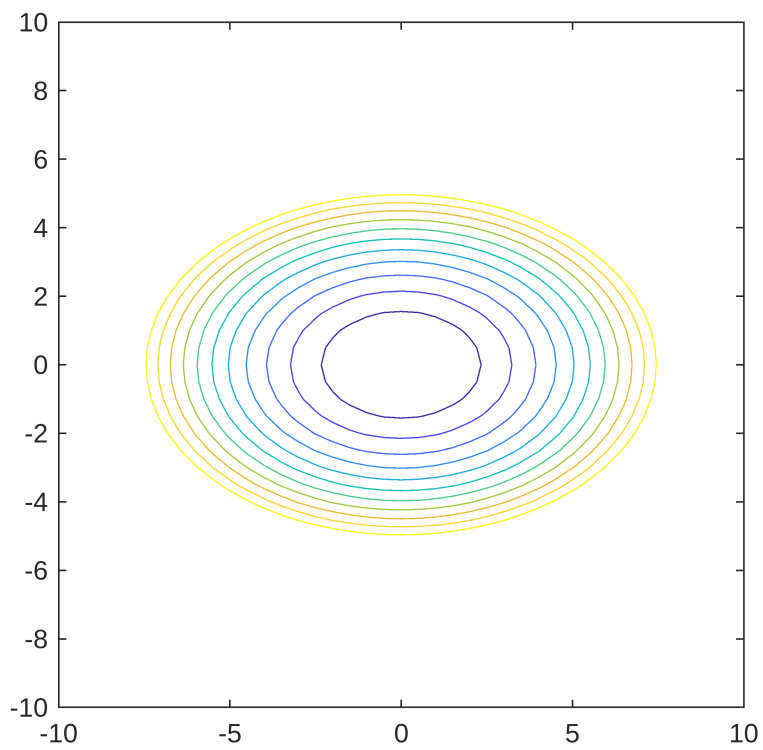
Question 7

```
[X, Y] = meshgrid(-5:0.1:5, -5:0.1:5);
Z = 1 ./ (abs(X) + abs(Y) + 0.5);
contour(X, Y, Z, 15)
```



Question 8

```
[x, y] = meshgrid(-10:0.5:10, -10:0.5:10);  
f = 4*x.^2 + 9*y.^2 - 72;  
v = -50:20:150;  
contour(x, y, f, v, 'ShowText', 'off')  
axis equal
```



```
% b
syms x y
f = 4*x^2 + 9*y^2 - 72;
grad_f = gradient(f, [x, y])
```

```
grad_f =
```

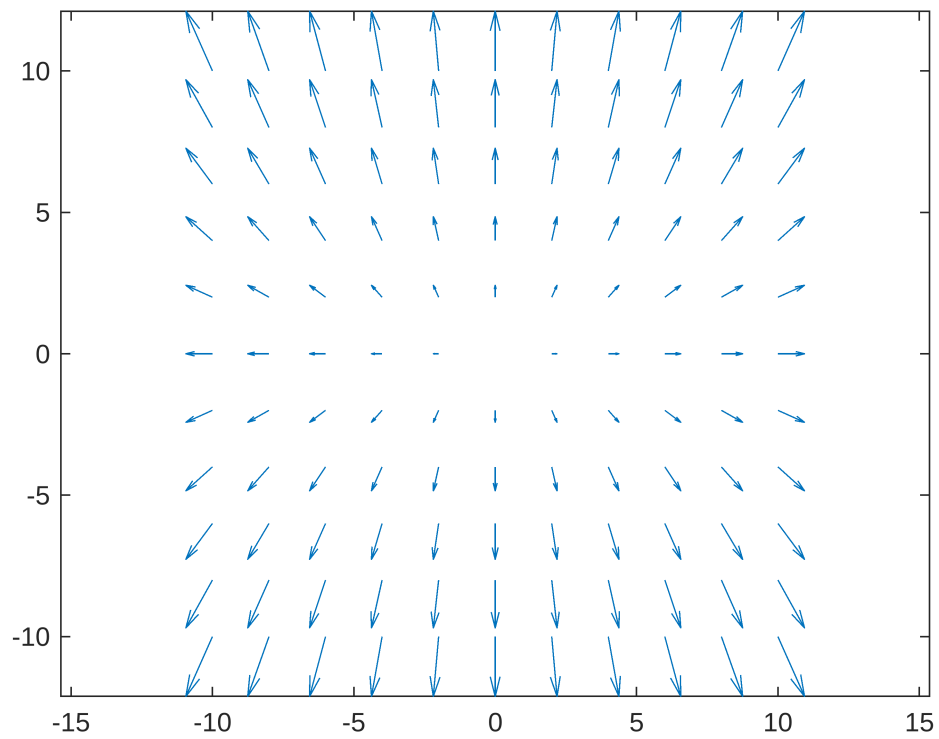
$$\begin{pmatrix} 8x \\ 18y \end{pmatrix}$$

```
hess_f = hessian(f, [x, y])
```

```
hess_f =
```

$$\begin{pmatrix} 8 & 0 \\ 0 & 18 \end{pmatrix}$$

```
% c
[x, y] = meshgrid(-10:2:10, -10:2:10);
u = 8*x;
v = 18*y;
quiver(x, y, u, v)
axis equal
```

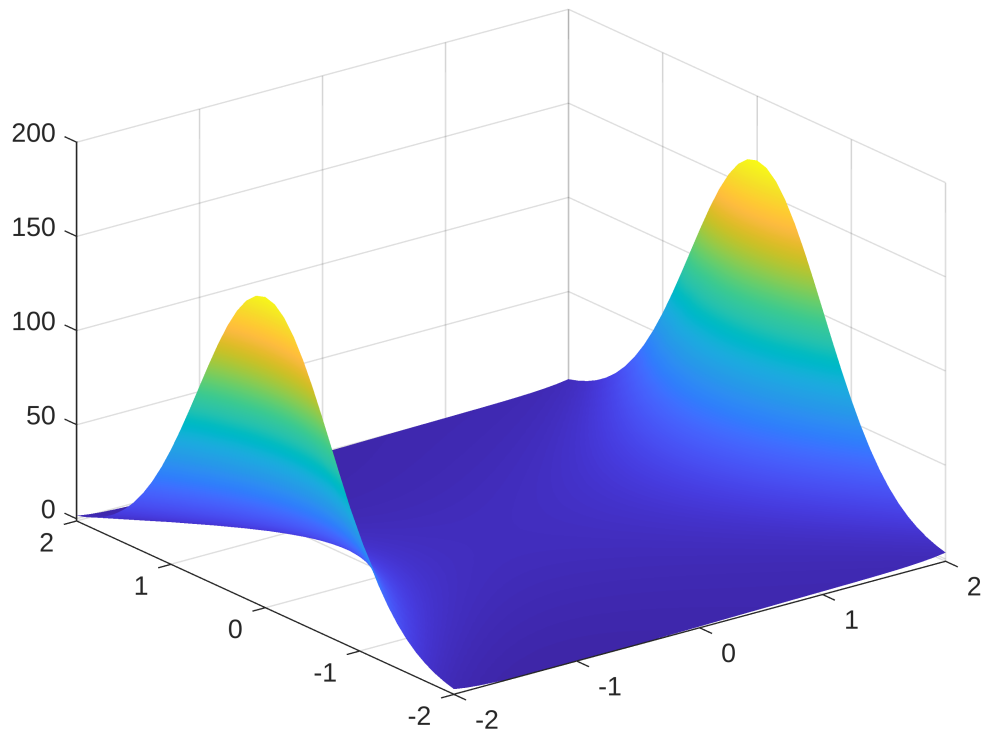


Question 9

% d) Surface plot and gradient vector field on level curves

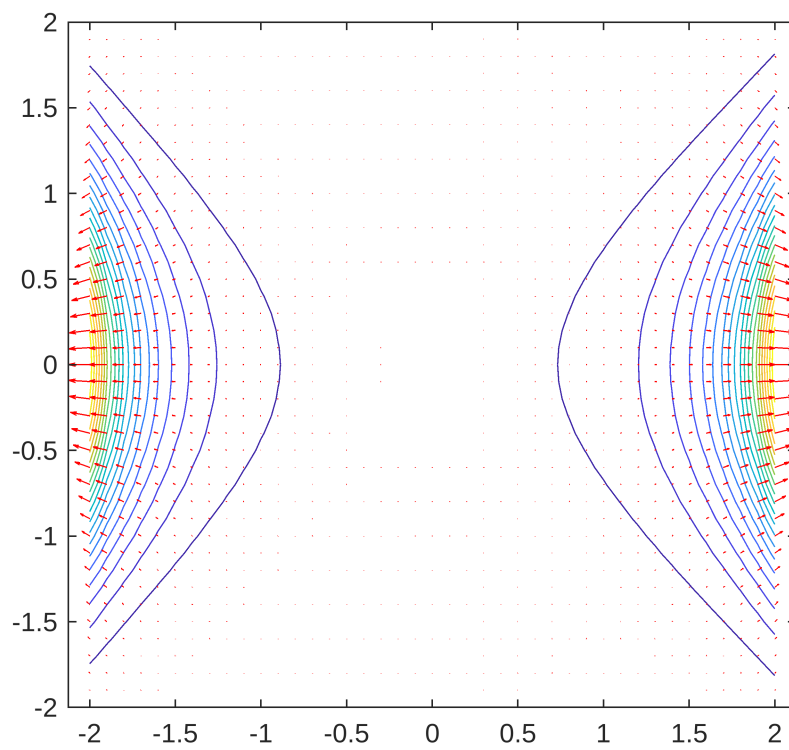
```
[x, y] = meshgrid(-2:0.1:2, -2:0.1:2);
f = sin(x) + cos(y) + 3*exp(x.^2 - y.^2);
```

```
figure
surf(x, y, f)
shading interp
```



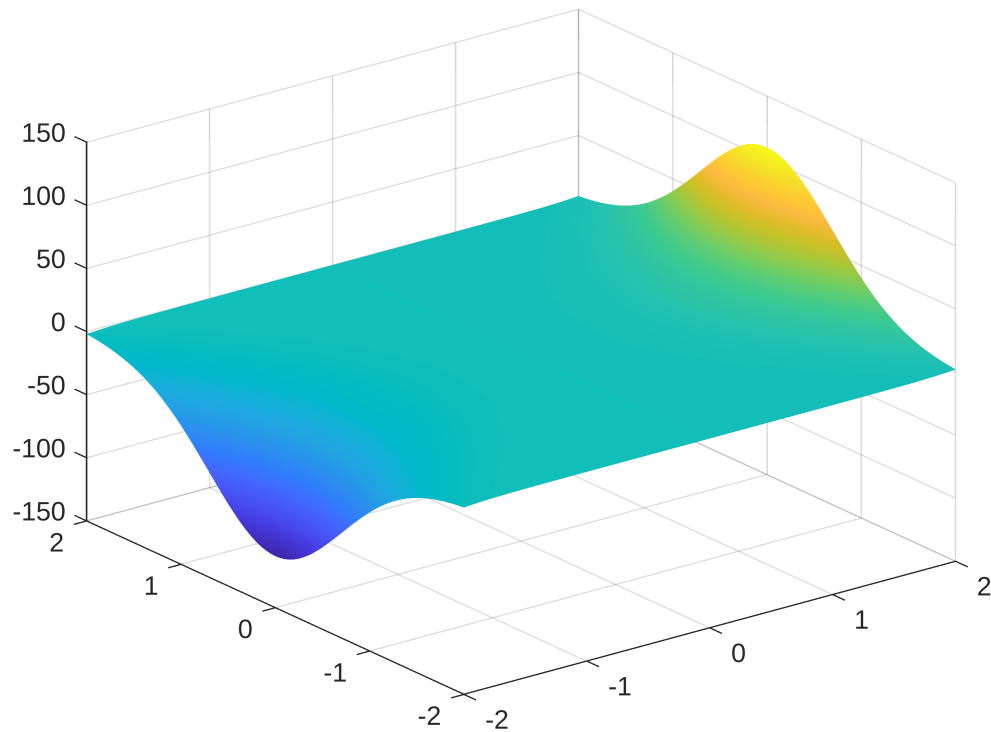
```
figure
contour(x, y, f, 20, 'ShowText', 'off')
hold on

[fx, fy] = gradient(f, 0.1, 0.1);
quiver(x, y, fx, fy, 'r')
axis equal
hold off
```



Question 10

```
% a
[x, y] = meshgrid(-2:0.05:2, -2:0.05:2);
f = x .* exp(x.^2 - y.^2);
surf(x, y, f)
shading interp
```



```
% b
syms x y
f = x * exp(x^2 - y^2);
grad_f = gradient(f, [x, y])
```

grad_f =

$$\begin{pmatrix} e^{x^2-y^2} + 2x^2 e^{x^2-y^2} \\ -2xy e^{x^2-y^2} \end{pmatrix}$$

```
hess_f = hessian(f, [x, y])
```

hess_f =

$$\begin{pmatrix} 4x^3 \sigma_1 + 6x \sigma_1 & -2y \sigma_1 - 4x^2 y \sigma_1 \\ -2y \sigma_1 - 4x^2 y \sigma_1 & 4xy^2 \sigma_1 - 2x \sigma_1 \end{pmatrix}$$

where

$$\sigma_1 = e^{x^2-y^2}$$

```
% c
x_val = 0; y_val = 0;
subs_hess = subs(hess_f, {x, y}, {x_val, y_val});
double_hess = double(subs_hess)
```

```
double_hess = 2x2
    0    0
    0    0
```

```
% d
eigvals = eig(double_hess);

if all(eigvals > 0)
    disp('Curved upward (local min)')
elseif all(eigvals < 0)
    disp('Curved downward (local max)')
elseif any(eigvals > 0) && any(eigvals < 0)
    disp('Saddle point (curved upward and downward)')
else
    disp('Flat or undetermined')
end
```

Flat or undetermined